

Nested Qualities and Local Responses: An Inherited Finite-Operator Construction

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Abstract

We give an explicit finite-operator construction exhibiting four nested qualities: coupled regional composition, an inherited law under a normalized scale transformation, preserved dimensionless spectral relationships, and an effective local response to unresolved regions. Every finite depth is covered by the same algebraic argument. The transformation acts on an indexed family of equal-depth windows, not by an unscaled dilation of a single fixed matrix. Admissible quantum states may differ between regions; common laws do not require identical occupations or behavior. Ordered Schur reduction retains the influence of a region's own surroundings, while time-domain elimination retains memory and initial-state information. These results reuse standard spectral and block-operator mathematics and connect the existing Nested-Space Cosmology formulation to a concrete model. They establish mathematical nested qualities within the stated finite class, without changing local quantum evolution. A separate gravitational incoming-gate calculation remains OPEN and is reported only as an application checkpoint.

1 Question and relation to the foundation

This article is a focused companion to the preserved Nested-Space Cosmology working manuscript, version 0.26.0 [1]. Its organizing relation is

$$\mathcal{T}_\Theta^* \mathbb{D}_\Theta = \mathbb{D}_\Theta. \quad (1)$$

The intended question is whether coupled regions can inherit a common law, while having different local states and retaining the influence of their surroundings in a reduced description. Here we specify a finite construction and a transformation for which these properties can be proved.

A region is a declared subspace of the finite system. Calling one region an inside and its complement an outside describes their relation in the chosen partition. It does not assign gauge charge or assert a particular horizon geometry. A common operator law is compatible with different initial states, occupations and correlations. A property of one region is therefore not assumed to be the property of every region.

The contribution is the connected construction and its precise interpretation. Schur elimination and unitary spectral covariance are established mathematics, not discoveries claimed here. We do not need to reconstruct published black-universe geometries to establish these finite properties. Such geometries provide separately scoped examples, with their own supporting sources [2]. Nor does the present theorem require a proof of eternal dynamics or closure of the numerical application in Appendix A.

2 An inherited family of coupled regions

Fix a finite regional dimension d , a Hermitian prototype H , a nonzero $d \times d$ coupling B , and a finite positive scale ratio Ω . Use orthonormal regional frames and common energy units with $\hbar = 1$. For an integer starting label m and a finite depth $N \geq 1$, let $\mathcal{H}_{m,N}$ be the direct sum of N copies of $\mathbb{C}^d$. The domain is the whole finite space. Define the Hermitian matrix $J_{m,N}$ by

$$(J_{m,N})_{jj} = \Omega^{m+j} H, \quad 0 \leq j < N, \quad (2)$$

$$(J_{m,N})_{j,j+1} = \Omega^{m+j} B, \quad (J_{m,N})_{j+1,j} = \Omega^{m+j} B^\dagger, \quad 0 \leq j < N-1, \quad (3)$$

with all other blocks zero. The nonzero link makes the construction coupled. The shared data are $\Theta = (H, B, \Omega)$; N specifies a boundary problem. The exterior remainder after the first region is removed is exactly the window $J_{m+1,N-1}$ when $N > 1$.

Proposition 1 (Inheritance under normalized relabeling). *Let $U_m : \mathcal{H}_{m,N} \rightarrow \mathcal{H}_{m+1,N}$ be the unitary map that relabels corresponding ordered regional frames. Then*

$$J_{m+1,N} = \Omega U_m J_{m,N} U_m^\dagger, \quad \Omega^{-1} U_m^\dagger J_{m+1,N} U_m = J_{m,N}. \quad (4)$$

In particular,

$$\text{spec}(\Omega^{-m} J_{m,N}) = \text{spec}(J_{0,N}). \quad (5)$$

Proof. Every nonzero block in (2) gains one factor Ω when m increases by one. The unitary U_m changes its label, not its matrix entries in corresponding frames. This proves (4) block by block. Iteration and invariance of spectrum under unitary conjugation give (5) for every integer m and finite N . $\square$

Equation (4) gives a concrete meaning to the pullback in (1): relabel the window and remove its declared scale. The invariant object is the indexed law/family. We do not assert that a nonzero finite matrix is unitarily equivalent to an unscaled dilation of itself. For fixed N , the frequency eigenvalues can be written $\omega_{m,k} = \Lambda_m \hat{\omega}_k(\Theta, N)$ with $\Lambda_m = \Omega^m$ in the chosen units. Isolated regional blocks obey the same statement without the N -dependence.

This does not make the dressed spectrum independent of surrounding regions. Changing N changes the boundary problem and can shift that spectrum. Likewise, the theorem does not determine a physical value of Ω or identify H, B with a particular curved-spacetime operator. Those are explicit inputs or separate identifications, not quantities inferred from notation.

3 Nested reduction retains the exterior influence

For a complex spectral parameter z with $\text{Im } z > 0$, all required resolvents exist: the finite Hamiltonian and its principal subblocks are Hermitian. For a retained block A and complement E , write

$$J = \begin{pmatrix} J_{AA} & V \\ V^\dagger & J_{EE} \end{pmatrix}. \quad (6)$$

Its retained resolvent is

$$[(zI - J)^{-1}]_{AA} = [zI - J_{AA} - \Sigma_A(z)]^{-1}, \quad \Sigma_A(z) = V(zI - J_{EE})^{-1} V^\dagger. \quad (7)$$

The unresolved region is absent from the retained basis but remains in its response. This is exact elimination, not removal of a physical influence.

Proposition 2 (Composition at every finite depth). *Starting at the final region, define*

$$S_{N-1}(z) = zI - \Omega^{m+N-1}H, \quad (8)$$

$$S_j(z) = zI - \Omega^{m+j}H - \Omega^{2(m+j)}BS_{j+1}(z)^{-1}B^\dagger. \quad (9)$$

Then $S_0(z)^{-1}$ is exactly the first regional block of $(zI - J_{m,N})^{-1}$. Grouping the exterior regions before eliminating them gives the same result.

Proof. The two-block inverse identity (7) establishes the terminal elimination. Apply it successively to the resulting complement; induction on N gives the claim. Alternatively eliminate the entire complement in one step. Both procedures equal the same block of the joined inverse, which proves agreement independently of the grouping. $\square$

For three blocks of $K = zI - J$, this statement is particularly visible:

$$K = \begin{pmatrix} A & -V & 0 \\ -V^\dagger & C & -W \\ 0 & -W^\dagger & F \end{pmatrix}, \quad S = A - V(C - WF^{-1}W^\dagger)^{-1}V^\dagger. \quad (10)$$

The outer contribution enters the middle response, which enters the retained response. Noncommuting products keep their displayed order.

Normalize S_j by Ω^{m+j} and set $x = z/\Omega^{m+j}$. The recurrence is

$$\Gamma_j(x) = xI - H - \Omega^{-1}B\Gamma_{j+1}(x/\Omega)^{-1}B^\dagger. \quad (11)$$

This is the repository's existing normalized chain. The factor $1/\Omega$ follows from the scale of the eliminated inverse; it is not an adjustable coupling. The proof covers arbitrary finite depth. No assumption about an infinite physical lifetime enters it.

4 The same law permits different regional states

Let C be any Hermitian fermionic one-particle covariance satisfying $0 \leq C \leq I$. Different regional blocks can have different occupations and cross-region correlations. Under the same static J ,

$$C(t) = e^{-iJt}C(0)e^{iJt}, \quad i\dot{C} = [J, C]. \quad (12)$$

Unitary conjugation preserves the interval $0 \leq C \leq I$, but does not require equality of regional blocks. For example $C_{\text{stat}} = I/2$ is stationary, whereas a nonuniform diagonal C_{var} with $[J, C_{\text{var}}] \neq 0$ evolves under the identical operator. Within that second state the regional occupations already differ. This is a statement about admissible states, not a simulation of stable stars or collapse.

Solving the exterior Schrödinger equation and substituting it into the retained equation yields

$$\begin{aligned} i\dot{\psi}_A &= J_{AA}\psi_A - i \int_0^t V e^{-iJ_{EE}(t-s)} V^\dagger \psi_A(s) ds \\ &\quad + V e^{-iJ_{EE}t} \psi_E(0). \end{aligned} \quad (13)$$

The initial exterior term must remain. Its covariance contains $V e^{-iJ_{EE}t} C_{EE}(0) e^{iJ_{EE}t} V^\dagger$; cross correlations also retain $C_{AE}(0)$. Thus the retarded operator and the state-dependent response are different objects. Equal operator laws need not produce equal measured histories.

Total energy is conserved for static J :

$$\frac{d}{dt} \text{Tr}(CJ) = -i \text{Tr}([J, C]J) = 0. \quad (14)$$

Regional energy can nevertheless be exchanged. With the existing allocation $h_A = (PJ + JP)/2$, the current is $i[J, h_A]$. This reuses finite quantum energy accounting and does not introduce a second independently chosen source. The Gaussian formulation can use established fermionic determinant identities [3]; a metric-source interpretation still requires the applicable common-action variation.

5 Verification, interpretation and scope

The analytical arguments establish the four qualities for the declared finite construction. A new small control record binds this proof to exact rational complex matrices: three two-dimensional regions, $\Omega = 3/2$ and $z = 1 + 2i$. Twelve exact predicates check the normalized shift, joined versus nested inverse, recurrence and distinct admissible states. Removing the normalization, reversing the noncommuting coupling or omitting the state contribution is detected. These are bounded integration checks; they do not replace the arbitrary-depth argument with a numerical depth scan.

The source and record are included under `paper/nested-quality-evidence` in the repository. The pinned source is lab commit `4b3c44d0c47a49a17f543ff19e0298770ac041e8`. The replay command is

```
python paper/nested-quality-evidence/scripts/
check_nsc_nested_qualities.py --check
```

It requires SymPy; the ordinary publication hash verifier does not. Earlier correspondence and recursion records are retained as authenticated references, with no claim that their whole gravitational dependency chain has become a new closed certificate.

This construction makes nested qualities mathematically explicit using standard local finite dynamics. It offers a precise way to describe an unresolved surrounding system through its local response. Identifying that response with cosmological dark matter or dark energy, or deriving numerical values of c and Λ , is not established by the Schur identity alone. Those identifications are not required for the result proved here.

The term *self-similar law* refers to the specified normalized transformation. It is not a measurement of a spatial fractal dimension of the Universe. An infinite geometric construction would need its own domains and limiting state; an eternal engine is neither assumed nor proved. Different regional states remain a feature of the construction throughout.

Author responsibility and assistance

Douglas Ek is the accountable author. Codex and Grok assisted with source review, algebraic controls, implementation and preparation of this manuscript. AI-generated descriptions were not treated as evidence. The full article and any submission remain subject to author review.

A Preserved gravitational application checkpoint

The separate source-fixed incoming calculation asks whether $C_\Sigma[g] = U_g C_{\text{up}} U_g^\dagger$, $\delta C_{\text{src}} = 0$ satisfies both incoming constraints on $I = S(1) + [0.12, 0.18]$ in $\delta r = \chi(s)[s w(z) + s^3 U(z)/6]$, $s =$

$T - T_\Sigma$. Its certificate remains **OPEN**. The recorded candidate has nodal maxima (9.64093×10^{-4} , 3.80960×10^{-4}), not certified continuous suprema. The numerical existence threshold is 3×10^{-11} with complete error control; it is not an acceptance threshold for the finite propositions above.

Three error components are enclosed in the preserved checkpoint; six remain unbounded. The finite operator theorem does not complete those bounds, select a physical radius history or change an old certificate’s meaning. The values below are generated from the original immutable records.

Component	N	β
Covered regions	8.91625×10^{-12}	3.88237×10^{-18}
Energy interpolation	4.83154×10^{-26}	3.78403×10^{-26}
Phase value	8.28992×10^{-14}	6.99570×10^{-14}
Field/space/time	OPEN	OPEN
Changed-history UV tail	OPEN	OPEN
Low/subgap	OPEN	OPEN
Upstream preparation	OPEN	OPEN
Between-node remainder	OPEN	OPEN
Remaining arithmetic	OPEN	OPEN

The source, chart, state and numerical assumptions of this application are separate from the finite theorem. Published curved-field constructions must be reused with their original domains [4]. No numerical campaign was rerun for the present synthesis, and no release or arXiv submission is implied by the completed finite argument.

References

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